

Study Important Questions for class 12 Chemistry

Chapter 3 - Chemical Kinetics

(Highly Expected Questions and Answer)

1. For the reaction $R \rightarrow P$, the concentration of a reactant changes from 0.03 M to 0.02M in 25 minutes. Calculate the average rate of reaction using units of time both in minutes and seconds.

Ans: Average rate of reaction $= -\frac{\Delta R}{\Delta t}$

$$-\frac{[R]_2 - [R]_1}{t_2 - t_1}$$

$$\Rightarrow -\frac{0.02 - 0.03}{25} = 4 \times 10^{-4} \text{ Mmin}^{-1}$$

Also, it can be expressed in seconds as:

$$\text{Average rate of reaction} = \frac{4 \times 10^{-4}}{60} = 6.67 \times 10^{-6}$$

2. In a reaction, $2A \rightarrow \text{Products}$, the concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes. Calculate the rate during this interval.

Ans: Average rate $= \frac{1}{2} \frac{\Delta [A]}{\Delta t}$

$$\frac{1}{2} \frac{[A]_2 - [A]_1}{t_2 - t_1}$$

$$= \frac{1}{2} \times \frac{0.4 - 0.5}{10}$$

$$= \frac{1}{2} \times \frac{1 - 0.1}{10}$$

$$0.005 \text{ mol L}^{-1}$$

$$= 5 \times 10^{-3} \text{ M min}^{-1}$$

3. For a reaction, $A + B \rightarrow \text{Product}$; the rate law is given by $r = k[A]^{\frac{1}{2}}[B]^2$. What is the order of the reaction?

Ans: The order of the reaction $= \frac{1}{2} + 2 = 2\frac{1}{2} = 2.5$

4. The conversion of molecules X to Y follows second order kinetics. If concentration of x is increased to three times how will it affect the rate of formation of Y?

Ans: Because the reaction X Y has second-order kinetics, the rate law equation will be $\text{Rate} = kC^2$, with $C = [x]$.

The rate law equation for the reaction X Y will be $\text{Rate} = k C^2$, with $C = [x]$ because it possesses second-order kinetics. So, $[x] = 3C \text{ mol L}^{-1}$

The rate equation is $\text{Rate} = K(3C)^2$

$$= 9(kC^2)$$

As a result, the reaction rate will increase by 9 times.

As a result, the rate at which Y is formed will grow by 9 times.

5. A first order reaction has a rate constant $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will 5g this reactant takes to reduce to 3g?

Ans: Initial amount = $[R]_0 = 5g$

Final concentration = $[R] = 3g$

Rate constant = $1.15 \times 10^{-3} \text{ s}^{-1}$

For a first order reaction $t = \frac{2.303}{k} \log \frac{[R]_0}{R}$

$$= \frac{2.303}{1.15 \times 10^{-3}} \log \frac{5}{3}$$

$$= \frac{2.303}{1.15 \times 10^{-3}} \times 0.2219$$

$$= 444.38 \text{ s or } 444 \text{ s}$$

6. Time required to decompose SO_2Cl_2 to half its initial amount is 60 minutes. If the decomposition is a first order reaction, calculate the rate constant of the reaction.

Ans: First order reaction $t_{\frac{1}{2}} = \frac{0.693}{k}$

$$t_{\frac{1}{2}} = 60 \text{ min}$$

$$k = \frac{0.693}{t_{\frac{1}{2}}}$$

$$= \frac{0.693}{60}$$

$$= 0.01155 \text{ min}^{-1}$$

$$= 1.155 \text{ min}^{-1}$$

$$k = 1.925 \times 10^{-4} \text{ s}^{-1}$$

7. What will be the effect of temperature on rate constant?

Ans: A 10^0 increase in temperature almost doubles the rate constant of a process. The Arrhenius equation, on the other hand, gives the exact temperature dependency of the rate of a chemical reaction.

$$k = Ae^{-E_a/RT}$$

Where A stands for the Arrhenius factor, also known as the frequency factor.

T stands for temperature.

The gas constant is R.

The activation energy is referred to as E_a .

8. The rate of the chemical reaction doubles for an increase of 10 K in absolute temperature from 298 K. Calculate E_a .

Ans: Given

$$T_1 = 298 \text{ K}$$

$$T_2 = (298 + 10) \text{ K}$$

We also know that when the temperature is raised by 10 degrees Celsius, the reaction rate doubles. As a result, we'll use the values $k_1 = k$ and $k_2 = 2k$.

$$R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\text{On substitution } \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\log \frac{2k}{k} = \frac{E_a}{2.303 \times 8.314} \left[\frac{10}{298 \times 308} \right]$$

$$E_a = \frac{2.303 \times 8.314 \times 298 \times 308 \times \log 2}{10}$$

$$= 52897.78 \text{ J mol}^{-1}$$

$$= 52.89 \text{ kJ mol}^{-1}$$

9. The activation energy for the reaction $2\text{HI}_{(g)} \rightarrow \text{H}_2 + \text{I}_{2(g)}$ is $209.5 \text{ kJ mol}^{-1}$ at 581 K . Calculate the fraction of molecules of reactants having energy equal to or greater than activation energy.

$$\text{Ans: } E_a = 209.5 \text{ kJ mol}^{-1} = 209500 \text{ J mol}^{-1}$$

$$T = 581 \text{ K}$$

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

The percentage of reactant molecules with energy equal to or greater than activation energy is now:

$$x = e^{E_a/RT}$$

$$\ln = -E_a / RT$$

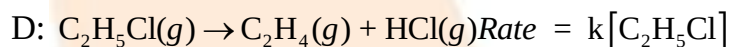
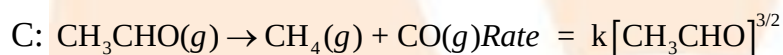
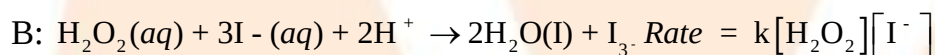
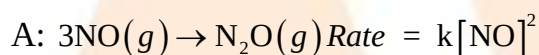
$$\log x = -\frac{E_a}{2.303RT}$$

$$\log x = \frac{209500 \text{ J mol}^{-1}}{2.303 \times 8.314 \times \text{J K}^{-1} \text{mol}^{-1} \times 581} = 18.8323$$

$$x = \text{Anti log}(-18.8323) = 1.47 \times 10^{-19}$$

NCERT Exercise

1. From the rate expression for the following reactions, determine their order of reaction and the dimensions of the rate constants.



Ans: Given

$$(i) . k = \frac{\text{Rate}}{[\text{H}_2\text{O}_2][\text{I}^-]}$$

Order of the reaction = 2

$$k = \frac{\text{Rate}}{[\text{NO}]^2}$$

$$\text{Therefore, Dimension of } k = \frac{\text{mol L}^{-1} \text{s}^{-1}}{(\text{mol L}^{-1})^2}$$

$$\frac{\text{mol L}^{-1} \text{s}^{-1}}{\text{mol}^2 \text{L}^{-2}} = \text{L mol}^{-1} \text{s}^{-1}$$

(ii) Given $k = [H_2O_2][I^-]$

Order of the reaction $k = \frac{\text{Rate}}{[H_2O_2][I^-]}$

Dimension of $k = \frac{\text{mol}^{-1}\text{s}^{-1}}{(\text{molL}^{-1})(\text{molL}^{-1})} = \text{L}^{-1}\text{mol}^{-1}\text{s}^{-1}$

(iii) Given $\text{rate} = k[CH_3CHO]^{3/2}$

Order of the reaction will be $\frac{3}{2}$

$k = \frac{\text{Rate}}{[CH_3CHO]^{3/2}}$

Dimension of $\frac{\text{mol}^{-1}\text{s}^{-1}}{(\text{mol L}^{-1})^{3/2}} = \frac{\text{mol}^{-1}\text{s}^{-1}}{\text{mol}^{3/2}\text{L}^{3/2}} = \text{L}^{1/2}\text{mol}^{-1/2}\text{s}^{-1}$

(iv) Given $\text{rate} = k[C_2H_5Cl]$

Order of the reaction $1k = \frac{\text{Rate}}{[C_2H_5Cl]}$

Dimension of $\frac{\text{mol L}^{-1}\text{s}^{-1}}{\text{mol L}^{-1}} = \text{s}^{-1}$

**2. For the reaction: $2A + B \rightarrow A_2B$ the rate = $k[A][B]^2$ with $k = 2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$.
Calculate the reaction's beginning rate when $[A] = 0.1 \text{ mol L}^{-1}$, $[B] = 0.2 \text{ mol L}^{-1}$
Calculate the rate of reaction after $[A]$, is reduced to 0.06 mol L^{-1}**

Ans: The initial rate of the reaction is $\text{Rate} = k[A][B]^2$

$$= (2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1})(0.1 \text{ mol L}^{-1})(0.2 \text{ mol L}^{-1})^2$$

$$= 8.0 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$$

The concentration of A responded when $[A]$ was lowered from 0.1 mol L^{-1} to 0.06 mol L^{-1} .

$$= (0.1 \times 0.06) \text{ mol L}^{-1} = 0.04 \text{ mol L}^{-1}$$

As a result, B concentration = $\frac{1}{2} \times 0.04 \text{ mol L}^{-1} = 0.02 \text{ mol L}^{-1}$

After then, the concentration of B responded.

$$[B] = 0.2 \times 0.02 \text{ mol L}^{-1} = 0.18 \text{ mol L}^{-1}$$

The rate of the reaction is given by once $[A]$ is lowered to 0.06 mol L^{-1} .

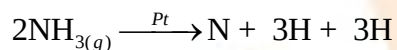
$$201\text{Rate} = k[A][B]^2$$

$$= (2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1})(0.06 \text{ mol L}^{-1})(0.18 \text{ mol L}^{-1})^2$$

$$= 3.89 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$$

3. The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if $k = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$?

Ans: The following equation represents the breakdown of NH_3 on a platinum surface.



2(g)

2(g)

2(g)

$$\text{Rate} = -\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$$

It is assumed, however, that the reaction is of zero order.

$$\text{So, Rate} = -\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$$

$$= 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\text{As a result, the rate of } \text{N}_2 \text{ production is } \frac{d[\text{N}_2]}{dt} = \frac{2.5 \times 10^{-4}}{2}$$

$$= 1.25 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

In addition, the rate of H_2 production is increasing.

$$\frac{d[\text{N}_2]}{dt} = \frac{3}{2} \times 2.5 \times 10^{-4}$$

$$= 3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

4. The decomposition of dimethyl ether leads to the formation of CH_4 , H_2 and CO and the reaction rate is given by $\text{Rate} = k = [\text{CH}_3\text{OCH}_3]^{3/2}$. The rate of reaction is followed by increase in pressure in a closed vessel, so the rate can also be expressed in terms of the partial pressure of dimethyl ether, i.e. $\text{Rate} = k = (P\text{CH}_3\text{OCH}_3)^{3/2}$. If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants?

Ans: In addition, the rate of H_2 production is increasing.

$$\text{Rate} = k = (P\text{CH}_3\text{OCH}_3)^{3/2}$$

$$k = \frac{\text{Rate}}{P\text{CH}_3\text{OCH}_3}$$

$$\text{Hence, the unit of rate constants } k = \frac{\text{bar min}^{-1}}{\text{bar}^{\frac{3}{2}}}$$

$$= \text{bar}^{-1/2} \text{ min}^{-1}$$

5. Mention the factors that affect the rate of a chemical reaction.

Ans: The rate of a reaction is influenced by several things.

(i) Reactant nature: The rate of the reaction is determined by the reactant's nature. Ionic compound reactions, for example, are faster than covalent compound reactions.

(ii) Reactant state: Solid reactions are sluggish, liquid reactions are quick, and gas reactions are very fast.

(iii) Temperature: The rate of reaction is heavily influenced by the temperature. Temperature raises the pace of reaction by 2-3 times for every 100°C increase in temperature.

$$\frac{r_T + 10}{r_T} = 2 - 3$$

The temperature coefficient is the name given to this proportion.

The average kinetic energy of reactant molecules increases as the temperature rises. As a result, the rate of collisions rises. The number of molecules with threshold energy grows as the temperature rises. As a result, the rate of reaction accelerates.

iv) Catalyst presence: The rate of reaction also is affected by the presence of a catalyst. Catalysts boost reaction rates by increasing reaction surface area, generating an unstable intermediate with the substrate, and providing a lower-activation-energy alternative path.

6. A reaction is second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is

(i) doubled

(ii) reduced to half?

Ans: Let the reactant concentration be $[A] = a$.

$$\text{Rate of reaction, } R = k[A]^2$$

$$= ka^2$$

(i) If the reactant concentration is doubled, $[A] = a$, the reaction rate is

$$R = k(2a)^2 = 4ka^2 = 4R$$

As a result, the reaction rate would increase by four times.

(ii) If the reactant's concentration is cut in half, $[A] = \frac{1}{2}a$ the rate of reaction will

$$\text{be } R = k\left(\frac{1}{2}a\right)^2$$

$$= \frac{1}{4} ka$$

$$= \frac{1}{4} R$$

As a result, the reaction rate would be lowered to $= \frac{1^{\text{th}}}{4}$

7. What is the effect of temperature on the rate constant of a reaction? How can this temperature effect on rate constant be represented quantitatively?

Ans: $k = Ae^{-E_a/RT}$

Where k is the rate constant,

A denotes the Arrhenius factor or frequency factor,

R denotes the gas constant,

T denotes the temperature, and E_a denotes the activation energy for the reaction.

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

k_1 = rate constant at temperature

T_1 k_1 = rate constant at temperature T_2

8. In a pseudo first order hydrolysis of ester in water, the following results were obtained:

t/s	0	30	60	90
[Ester] mol L ⁻¹	0.55	0.31	0.17	0.085

(i) Calculate the average rate of reaction between the time interval 30 to 60 seconds.

(ii) Calculate the pseudo first order rate constant for the hydrolysis of ester

Ans:

(i) Average reaction time between 30 and 60 seconds = $\frac{d[\text{Ester}]}{dt}$

$$\frac{0.31 - 0.17}{60 - 30} = \frac{0.14}{30}$$

$$= 4.67 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

(ii) pseudo first order reaction

$$k = \frac{2.303}{t} \log \frac{[\text{Ester}]_0}{[\text{Ester}]}$$

When $t = 30\text{s}$

$$k = \frac{2.303}{30} \times \log \left(\frac{0.55}{0.31} \right) = 1.91 \times 10^{-2} / \text{s}$$

When $t = 60\text{s}$

$$k = \frac{2.303}{60} \times \log\left(\frac{0.55}{0.17}\right) = 1.96 \times 10^{-2} / s$$

When $t = 90s$

$$k = \frac{2.303}{90} \times \log\left(\frac{0.55}{0.085}\right) = 2.07 \times 10^{-2} / s$$

$$\text{Average rate constant } k = \frac{k_1 + k_2 + k_3}{3} = 1.98 \times 10^{-2} / s$$

9. A reaction is first order in A and second order in B.

i. Write the differential rate equation.

ii. How is the rate affected on increasing the concentration of three times?

iii. How is the rate affected when the concentrations of both A and B are doubled

Ans:

(i) The differential rate equation will be the following: $\frac{d[R]}{dt} = k[A][B]^2$

(ii) If B's concentration is three times higher, then $\frac{d[R]}{dt} = k[A][3B]^2 = 9.k[A][B]^2$

As a result, the reaction rate will rise by 9 times.

(iii) When both A and B concentrations are doubled,

$$\frac{d[R]}{dt} = k[A][B]^2$$

$$= k[2A][2B]^2$$

$$= 8.k [A] [B]^2$$

As a result, the reaction rate will rise by eight times.

10. In a reaction between A and B the initial rate of reaction (r_0) was measured for different initial concentrations of A and B as given below:

A/mol L ⁻¹	0.20	0.20	0.04
B/mol L ⁻¹	0.30	0.10	0.05
r_0 mol L ⁻¹ s ⁻¹	5.07×10^{-5}	5.07×10^{-5}	1.43×10^{-4}

What is the order of the reaction with respect to A and B?

Ans:

Let the reaction order be x with respect to A and y with respect to B. Therefore,

$$r_0 = k[A]^x [B]^y$$

$$5.07 \times 10^{-5} = k[0.20]^x [0.30]^y \dots (i)$$

$$5.07 \times 10^{-5} = k[0.20]^x [0.10]^y \dots (ii)$$

$$1.43 \times 10^{-5} = k[0.40]^x [0.05]^y \dots (iii)$$

$$\frac{5.07 \times 10^{-5}}{5.07 \times 10^{-5}} = \frac{k[0.20]^x [0.30]^y}{k[0.20]^x [0.10]^y}$$

$$I = \frac{[0.30]^y}{[0.10]^y}$$

$$\left[\frac{0.30}{0.10}\right]^x \left[\frac{0.30}{0.10}\right]^y$$

$$y = 0$$

We get equation (ii) by dividing equation (ii) by equation (ii).

$$\frac{1.43 \times 10^{-4}}{5.07 \times 10^{-5}} = \frac{k [0.40]^x [0.05]^y}{k [0.20]^y [0.30]^y}$$

$$\frac{1.43 \times 10^{-4}}{5.07 \times 10^{-5}} = \frac{k [0.40]^x}{k [0.20]^y}$$

$$0.05]^y = [0.30]^y = 1$$

$$2.821 = 2^x$$

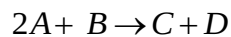
$$\log 2.821 = x \log 2$$

$$x = \frac{\log 2.821}{\log 2}$$

$$1.496 = 1.5 (\text{approximately})$$

As a result, the reaction's order regarding A is 1.5 and zero with respect to B.

11. The following results have been obtained during the kinetic studies of the reaction:



Experiment	A/mol L ⁻¹	B/mol L ⁻¹	Initial rate of formation of D/mol L ⁻¹ min ⁻¹
t			

I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.3×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Determine the rate law and the rate constant for the reaction

Ans: Let the reaction order be x for reaction with respect to A and y for reaction with respect to B. As a result, the reaction rate is given by, $Rate = k[A]^x[B]^y$

$$6.0 \times 10^{-3} = k[0.1]^x[0.1]^y \text{ (i)}$$

$$7.2 \times 10^{-2} = k[0.3]^x[0.2]^y \text{ (ii)}$$

$$2.88 \times 10^{-1} = k[0.3]^x[0.1]^y \text{ (iii)}$$

$$2.40 \times 10^{-2} = k[0.4]^x[0.1]^y \text{ (iv)}$$

When we divide equation (iv) by I we get,

$$\frac{2.40 \times 10^{-2}}{6.0 \times 10^{-3}} = \frac{k[0.4]^x[0.1]^y}{k[0.1]^x[0.1]^y}$$

$$4 = \frac{[0.4]^x}{[0.1]^x}$$

$$4 = \left(\frac{0.4}{0.1}\right)^x$$

$$x = 1$$

Divide (iii) by (i)

$$\frac{2.88 \times 10^{-1}}{7.2 \times 10^{-2}} = \frac{k[0.3]^x [0.4]^y}{k[0.3]^x [0.2]^y}$$

$$4 = \left(\frac{0.4}{0.2}\right)^y$$

$$4 = 2^y$$

$$2^2 = 2^y$$

$$y = 2$$

Rate law

$$\text{Rate} = k [A] [B]^2$$

$$k = \frac{\text{Rate}}{[A][B]^2}$$

From experiment I we understand

$$k = \frac{6.0 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.1 \text{ mol L}^{-1})(0.1 \text{ mol L}^{-1})^2}$$

$$= 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

From experiment II we understand

$$k = \frac{2.88 \times 10^{-1} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.3 \text{ mol L}^{-1})(0.4 \text{ mol L}^{-1})^2} = 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

From experiment III we understand

$$k = \frac{2.40 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}}{(0.4 \text{ mol L}^{-1})(0.1 \text{ mol L}^{-1})^2} = 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

Hence, the rate constant $k = 6.0 \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$

12. The reaction between A and B is first order with respect to A and zero order with respect to B. Fill in the blanks in the following table:

Experiment	A/mol L ⁻¹	B/mol L ⁻¹	Initial rate/mol L ⁻¹ min ⁻¹
I	0.1	0.1	2.0×10^{-2}
II	----	0.2	4.0×10^{-2}
III	0.4	0.4	----
IV	----	0.2	2.0×10^{-2}

Ans: With respect to A, the given reaction is first order, whereas with respect to B, it is zero order.

The rate of the reaction $\text{Rate} = k[A]^1[B]^0$

$$\text{Rate} = k = [A]$$

From 1st experiment,

$$2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = k(0.1 \text{ mol L}^{-1})$$

$$k = 0.2 \text{ min}^{-1}$$

From 2nd experiment,

$$4.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = k(0.2 \text{ mol L}^{-1})$$

$$[A] = 0.2 \text{ mol L}^{-1}$$

From 3rd experiment,

$$\begin{aligned} \text{Rate} &= 0.2 \text{ min}^{-1} \times 0.4 \text{ mol L}^{-1} \\ &= 0.08 \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

From 4th experiment

$$2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1} = 0.2 \text{ min}^{-1} [A]$$

$$[A] = 0.1 \text{ mol L}^{-1}$$

13. Calculate the half-life of a first order reaction from their rate constants given below:

(i) 200 s^{-1} (ii) 2 min^{-1} (iii) 4 years^{-1}

Ans:

(i) Half life $t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{200 \text{ s}^{-1}} = 0.346 \times 10^{-2}$

(ii) Half life $t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{2 \text{ min}^{-1}} = 0.346 \text{ min}$

(iii) Half life $t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{4 \text{ years}^{-1}} = 0.173 \text{ years}$

14. The half-life for radioactive decay of ^{14}C is 5730 years. An archeological artifact containing wood had only 80% of the ^{14}C found in a living tree. Estimate the age of the sample.

Ans: The dynamics of radioactive decay are first order. As a result, the decay constant

$$\text{Decay constant } k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730 \text{ year}} = 1.209 \times 10^{-4} / \text{ years}$$

The rate of counts is proportional to the number of C-14 atoms in the sample.

$$N_0 = 100, N = 80$$

$$\text{The age of the sample } t = \frac{2.303}{k} \log \left(\frac{N_0}{N} \right)$$

$$t = \frac{2.303}{1.209 \times 10^{-4}} \times \log \left(\frac{100}{80} \right) = 1846 \text{ years}$$

Therefore, the sample age is 1846 years.

15. The experimental data for decomposition of N_2O_5 [$2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$] in gas phase at 318K are given below:

t/s	0	400	800	1200	1600	2000	2400	1800	3200
$10^{-2} \times \text{N}_2\text{O}_5$ mol L ⁻¹	1.63	1.36	1.14	0.93	0.78	0.64	0.53		

(i) Plot $\log[\text{N}_2\text{O}_5]$ against t.

(ii) Find the half-life period for the reaction.

(iii) Draw a graph between $\log[N_2O_5]$ and t .

(iv) What is the rate law?

(v) Calculate the rate constant.

Calculate the half-life period from k and compare it with (ii)

Ans:

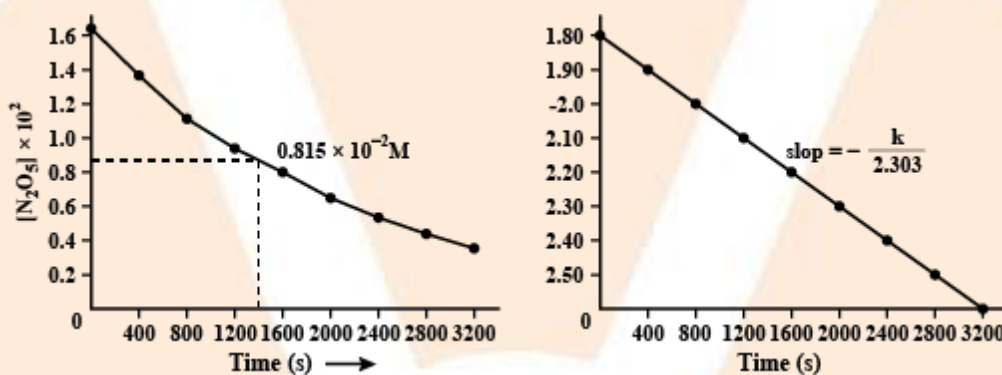


Image: A graph between
 $\log[N_2O_5]$ and t .

The period that corresponds to the concentration

$$\frac{1.630 \times 10^{-2}}{2} \text{ mol L}^{-1} = 0.815 \times 10^{-2} \text{ mol L}^{-1} \text{ is half-life.}$$

Observing the graph, the half life is 1440s.

(iii)

t(s)	$10^2 \times [N_2O_5] \text{ mol L}^{-1}$	$\log[N_2O_5]$
------	---	----------------

0	1.63	-1.79
400	1.36	-1.87
800	1.14	-1.94
1200	0.93	-2.03
1600	0.78	-2.11
2000	0.64	-2.19
2400	0.53	-2.28
2800	0.43	-2.37
3200	0.35	-2.46

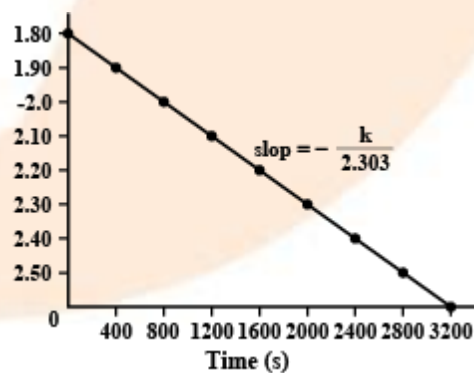
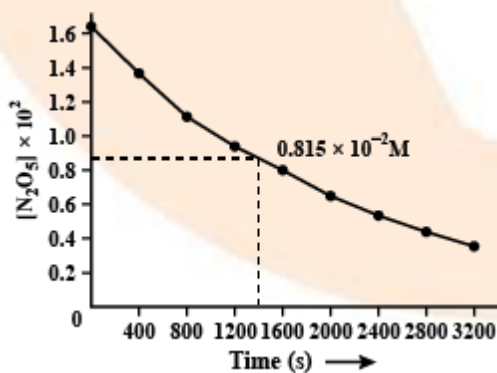


Image: A straight-line plot of log

N_2O_5 v/s time

The provided reaction is of first order, as evidenced by the straight-line plot of $\log N_2O_5$ v/s t. As a result, the reaction's rate law is $Rate = k[N_2O_5]$

(v) From the plot N_2O_5 v/s t, we get

$$= \frac{2.46 - (-1.79)}{3200 - 0} = \frac{-0.67}{3200}$$

$$= 4.82 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

(vi) Half life given by

$$= \frac{2.46 - (-1.79)}{3200 - 0} = \frac{-0.67}{3200}$$

$$= 4.82 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$= 1438 \text{ s}$$

The value of $t_{\frac{1}{2}}$ computed from k is extremely close to the value acquired from graph.

16. Rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $1/16^{\text{th}}$ value?

Ans: $t = \frac{2.303}{k} \log \frac{[R]}{[R]}$

$$= \frac{2.303}{60 \text{ s}^{-1}} \log \frac{1}{16}$$

$$= \frac{2.303}{60\text{s}^{-1}} \log 16$$

$$= 4.62 \times 10^{-2} \text{s}$$

17. During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If 1 μg of ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically?

$$\text{Ans: } k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{28.1} \text{y}^{-1}$$

Knowing

$$t = \frac{0.303}{k} \log \frac{[R]}{[R]}$$

$$t = \frac{0.303}{\frac{0.693}{28.1}} \log \frac{1}{[R]} 10$$

$$= \frac{0.303}{\frac{0.693}{28.1}} (-108[R]) \log [R] - \frac{10 \times 0.303}{2.303 \times 28.1} [R]$$

$$= \text{anti log}(-0.1071) = 0.7814 \mu\text{g}$$

Hence, 0.7814 μg of ^{90}Sr will remain after 10 years.

$$\text{Repeating } k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{28.1} \text{y}^{-1}$$

$$\text{Knowing } t = \frac{0.303}{k} \log \frac{[R]}{[R]}$$

$$60 = \frac{0.303}{\frac{0.693}{28.1}} \log \frac{1}{[R]}$$

$$\log[R] = -\frac{10 \times 0.303}{2.303 \times 28.1}$$

$$[R] = \text{anti log}(-0.6425)$$

$$= 0.2278 \mu\text{g}$$

Thus, 0.2278 μg of ^{90}Sr will remain after 60 years

18. For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction

Ans: The time required for 99 percent completion of a first order reaction is

$$\begin{aligned} t_1 &= \frac{2.303}{k} \log \frac{100}{100-99} \\ &= \frac{2.303}{k} \log 100 \end{aligned}$$

The time required for 90 percent completion of a first order reaction is

$$\begin{aligned} t_2 &= \frac{2.303}{k} \log \frac{100}{100-90} \\ &= \frac{2.303}{k} \log 10 \\ &= \frac{2.303}{k} \end{aligned}$$

$$t_1 = 2t_2$$

As a result, the time required to complete a first order reaction at 99 percent is twice the time required to complete the reaction at 90%.

19. A first order reaction takes 40 min for 30% decomposition. Calculate $t_{\frac{1}{2}}$

Ans: 1st order reaction

$$t_2 = \frac{2.303}{k} \log \frac{[R]}{[R]}$$

$$k = \frac{2.303}{40 \text{ min}} \log \frac{100}{100 - 30}$$

$$= \frac{2.303}{40 \text{ min}} \log \frac{10}{7}$$

$$= 8.918 \times 10^{-3} \text{ min}^{-1}$$

$t_{\frac{1}{2}}$ of the decomposition reaction is

$$t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{8.918 \times 10^{-3}} \text{ min}$$

$$= 77.7 \text{ min}$$

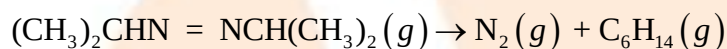
20. For the decomposition of azoisopropane to hexane and nitrogen at 543 K, the following data are obtained.

t(sec)	P(mm of Hg)
0	35.0

360	54.0
720	63.0

Calculate the rate constant

Ans: The following equation represents the breakdown of azoisopropane to hexane and nitrogen at 54.3 k.



At $t = 0$ P_0

$$t = tP_0 - p$$

Total pressure, $(P_t) = (P_0 - p) + p + p$

$$P_t = P_0 + p$$

when the value of p is substituted for the pressure of the reactant at time t

$$= P_0 - p$$

$$= P_0 - (P_t - P_0)$$

$$= 2P_0 - P_t$$

$$\text{1st order reaction } k = \frac{2.303}{t} \log \frac{P}{2P_0 - P_t}$$

$$t = 360 \text{ s ,}$$

$$k = \frac{2.303}{360} \log \frac{35.0}{2 \times 35.0 - 54.0}$$

$$= 2.175 \times 10^{-3} \text{ s}^{-1}$$

$$\text{when } t = 720\text{s}, k = \frac{2.303}{720} \log \frac{35.0}{2 \times 35.0 - 63.0} 10^{-3} \text{ s}^{-1}$$

Average value of rate constant

$$k = \frac{(2.175 \times 10^{-3}) + (2.235 \times 10^{-3})}{2} \text{ s}^{-1}$$

$$= 2.20 \times 10^{-3} \text{ s}^{-1}$$

21. The following data were obtained during the first order thermal decomposition of SO_2Cl_2 at a constant volume $\text{SO}_2\text{Cl}_2(g) \rightarrow \text{SO}_2(g) + \text{Cl}_2(g)$

Experiment	Time/ s^{-1}	Pressure/atm
1	0	0.5
2	100	0.6

Calculate the rate of the reaction when total pressure is 0.65 atm.

Ans: The following equation represents the thermal breakdown of SO_2Cl_2 at a constant volume.



At $t = 0 P_0$

$$t = t P_0 - p$$

Total pressure, $(P_t) = (P_0 - p) + p + p$

$$P_t = P_0 + p$$

when the value of p is substituted for the pressure of the reactant at time t

$$= P_0 - p$$

$$= P_0 - (P_t - P_0)$$

$$= 2P_0 - P_t$$

$$k = \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_t}$$

$$k = \frac{2.303}{100s} \log \frac{0.5}{2 \times 0.5 - 0.6}$$

when $t = 100s$

$$K = 2.231 \times 10^{-3} s^{-1}$$

When $P_t = 0.65 \text{ atm}$,

As a result, the total pressure of SO_2Cl_2 at time t is 0.65 atm.

$$2 P_{SOCl} = 2 P_0 - P_t$$

$$= 2 \times 0.50 - 0.65$$

$$= 0.35 \text{ atm}$$

As a result, when total pressure is 0.65 atm, the rate of equation is

$$\text{Rate} = k (P_{SOCl})$$

$$= (2.33 \times 10^{-3} s^{-1})(0.354 \text{ atm}) = 7.8 \times 10^{-4} \text{ atm s}^{-1}$$

22. The rate constant for the decomposition of N_2O_5 at various temperatures is given below:

$T/^{\circ}C$	0	20	40	60	80
$10^5 \times k/s^{-1}$	0.0787	0.0787	0.0787	178	2140

Draw a graph between $\ln k$ and $1/T$ and calculate the values of A and E_a . Predict the rate constant at 30° and $50^{\circ}C$.

Ans: From the given data, we get

$T/^{\circ}C$	0	20	40	60	80
$1/T / K^{-1}$	$0.2733.66 \times 10^{-3}$	$0.20933.41 \times 10^{-3}$	$0.403133.19 \times 10^{-3}$	$0.60333.3 \times 10^{-3}$	$0.80353.283 \times 10^{-3}$
$10^5 \times k/s^{-1}$	0.0787	4.075	25.7	178	2140
$\ln k$	-7.147	-4.075	-1.359	-0.577	3.063

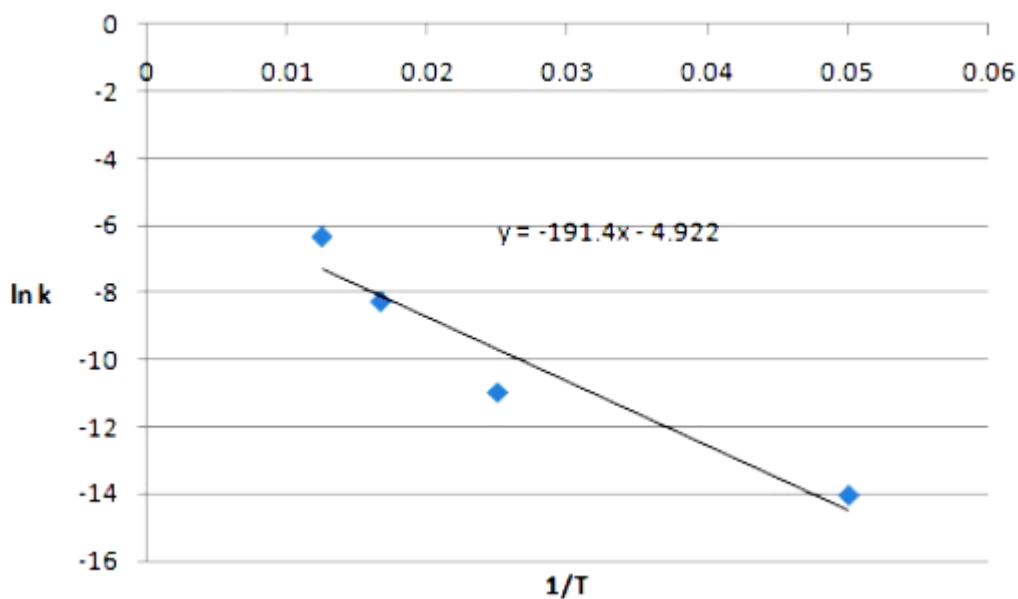


Image: A graph between $\ln k$ and $1/T$

Slope of the line $\frac{y_2 - y_1}{x_2 - x_1} = 12.301k$

Arrhenius equation,

$$\text{Slope} = -\frac{E_a}{R} \rightarrow E_a = -\text{slope} \times R = -(-12.301K) \times (8.314 JK^{-1} \text{mol}^{-1}) = 102.27 \text{ K J mol}^{-1}$$

$$\text{Repeating } \ln k = \ln A - \frac{E_a}{ET}$$

$$\ln A = \ln k + \frac{E_a}{ET}$$

Since, $T = 273K$, $\ln k = -7.147$

$$\text{So, } \ln A = -7.147 + \frac{102.27 \times 10^3}{8.314 \times 273} = 37.911A = 2.91 \times 10^6$$

$$T = 30 + 273K = 303K$$

$$\frac{I}{T} = 0.0033K = 3.3 \times 10^{-3} K$$

$$\text{At, } \frac{I}{T} = 0.0033K = 3.3 \times 10^{-3} K$$

$$\ln k = -2.8$$

$$k = 6.08 \times 10^{-2} s^{-1}$$

$$\frac{I}{T} = 0.0031K = 3.1 \times 10^{-3} K$$

$$\ln k = -0.5$$

$$k = 0.607 s^{-1}$$

$$\frac{I}{T} = 3.1 \times 10^{-3}$$

$$T = 50 + 273K = 323K$$

23. The rate constant for the decomposition of hydrocarbons is $2.418 \times 10^{-5} s^{-1}$ at 546 K. If the energy of activation is 179.9 kJ/mol, what will be the value of pre-exponential factor.

Ans: Arrhenius equation

$$\begin{aligned} K &= Ae^{-E_a/RT} \quad \ln k = \ln A - \frac{E_a}{RT} \quad \log k = \log A - \frac{E_a}{2.303RT} \\ &= \log (2.418 \times 10^{-5} s^{-1}) + \frac{179.9 \times 10^3 Jmol^{-1}}{2.303 \times 8.314 JKmol^{-1} \times 546K} \\ &= (0.3835 - 5) + 17.2082 = 12.5917 \end{aligned}$$

$$A = 3.912 \times 10^{12} \text{ s}^{-1}$$

24. Consider a certain reaction $A \rightarrow \text{Products}$ with $k = 2.0 \times 10^{-2} \text{ s}^{-1}$. Calculate the concentration of A remaining after 100 s if the initial concentration of A is 1.0 mol L^{-1} .

Ans: $k = 2.0 \times 10^{-2} \text{ s}^{-1}$, $t = 100 \text{ s}$, $[A]_0 = 1.0 \text{ mol L}^{-1}$

The given reaction is a first order reaction since the unit of k equals s^{-1}

$$k = \frac{2.303}{t} \log \frac{[A]}{[A]_0} \quad 2.0 \times 10^{-2} \text{ s}^{-1} = \frac{2.303}{100 \text{ s}} \log \frac{1.0}{[A]} \quad 2.0 \times 10^{-2} \text{ s}^{-1} = \frac{2.303}{100 \text{ s}} (-\log [A])$$

$$-\log [A] = \frac{2.0 \times 10^{-2} \times 100}{2.303}$$

$$[A] = \text{anti log} \left(\frac{2.0 \times 10^{-1} \times 100}{2.303} \right)$$

$$= 0.135 \text{ mol L}^{-1}$$

As a result, the concentration of A left is 0.135 mol L^{-1}

25. Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law with $t_{1/2} = 3 \text{ hours}$. What fraction of sample of sucrose remains after 8 hour?

hour?

Ans: 1st order reaction

$$k = \frac{2.303}{t} \log \frac{[R_0]}{[R]}$$

$$k = \frac{2.303}{t} \log \frac{[R_0]}{[R]}$$

$$t_{\frac{1}{2}} = 3 \text{ hours}$$

$$k = \frac{0.693}{t_{\frac{1}{2}}}$$

$$\text{So, } k = \frac{0.693}{3} = 0.231h^{-1}$$

$$= 0.231h^{-1} = \frac{2.303}{8h} \log \frac{[R]_0}{[R]}$$

$$\log \frac{[R]_0}{[R]} = \frac{0.231h^{-1} \times 8h}{2.303}$$

$$\frac{[R]_0}{[R]} = \text{anti log}(0.8024)$$

$$\frac{[R]_0}{[R]} = 6.3445$$

$$\frac{[R]_0}{[R]} = 0.1576$$

$$= 0.158$$

As a result, after 8 hours, 0.158 percent of the sucrose sample remains.

26. The decomposition of hydrocarbon follows the equation

$$k = (4.5 \times 10^{11} \text{ s}^{-1}) e^{-28000 K / T} . \text{ Calculate } E_a$$

Ans: From the given equation $k = (4.5 \times 10^{11} \text{ s}^{-1}) e^{-28000 K / T}$ (i)

Arrhenius equation ,

$$k = Ae^{-E/RT} \text{ (ii)}$$

From equation (i) and (ii) we get

$$\frac{E_a}{RT} = \frac{28000K}{T}$$

$$E_a = R \times 28000K$$

$$= 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 28000k$$

$$= 232792 \text{ J mol}^{-1}$$

$$= 232.792 \text{ kJ mol}^{-1}$$

27. The rate constant for the first order decomposition of H_2O_2 is given by the following equation: $\log k = 14.34 - 1.25 \times 10^4 K / T$. Calculate E_a for this reaction and at what temperature will its half-period be 256 minutes?

Ans: The expression for the rate constant is as follows:

$$\log k = 14.34 - 1.25 \times 10^4 K / T \dots (i)$$

Comparing it with Arrhenius equation, we get-

$$\log k = \log A - \frac{E_a}{2.303RT}$$

Therefore, $\frac{E_a}{2.303R} = 1.25 \times 10^4$

$$E_a = 1.25 \times 10^4 \times 2.303 \times 8.314$$

The activation energy = $E_a = 239339 \text{ J / mol} = 239.339 \text{ kJ / mol}$

Half life period, $t_{1/2} = 256 \text{ min} = 256 \times 60 \text{ sec}$

$$k = \frac{0.693}{t_{1/2}}$$

$$k = \frac{0.693}{256 \times 60 \text{ sec}}$$

$$k = 4.51 \times 10^{-5} / \text{s}$$

Substitute in equation (i), we get-

$$\log 4.51 \times 10^{-5} = 14.341.25 \times 10^4 \text{ K / T}$$

$$-4.35 = 14.341.25 \times 10^4 \text{ K / T}$$

$$T = 669 \text{ K}$$

Hence, the temperature at which the half-life period is 256 minutes is 669K.

28. The decomposition of A into product has value of k as $4.5 \times 10^3 \text{ s}^{-1}$ at 10°C and energy of activation 60 kJ mol^{-1} . At what temperature would k be $1.5 \times 10^4 \text{ s}^{-1}$?

Ans: It is given that the decomposition of A into product has value of

$$k = 4.5 \times 10^3 \text{ s}^{-1} \text{ at } 10^\circ \text{C}.$$

$$\log \left(\frac{K'}{k} \right) = \frac{E_a}{2.303R} \left[\frac{T' - T}{TT'} \right]$$

$$\log \left(\frac{1.5 \times 10^4}{4.5 \times 10^3} \right) = \frac{60 \times 10^3}{2.303 \times 8.314} \left[\frac{T' - 283}{283T'} \right]$$

$$0.5228 = 3132.62 \left[\frac{T' - 283}{283T'} \right]$$

$$T' - 283 = 0.0472T'$$

$$T' = 297.02 \text{ K}$$

$$T' = 297.02 \text{ K} - 273 = 24.02^\circ \text{ C}$$

29. The time required for 10% completion of a first order reaction at 298 K is equal to that required for its 25% completion at 308 K. If the value of A is $4 \times 10^{10} \text{ s}^{-1}$, Calculate k at 318 K and E_a

Ans: 1st order reaction

$$t = \frac{2.303}{k} \log \frac{a}{a-x}$$

$$\text{at } 298 \text{ K}, t = \frac{2.303}{k} \log \frac{100}{90}$$

$$= \frac{0.1054}{k}$$

$$- \frac{2.303}{k} \log \frac{100}{75}$$

$$\text{At } 308 \text{ K} = \frac{2.2877}{k}$$

Regarding question

$$= \frac{0.1054}{k} \log \frac{0.2877}{k}$$

$$\log \frac{k_1}{k} = 2.7296$$

From Arrhenius equation, we get

$$\log, \log \frac{k_1}{k} = \frac{E_a}{2.303R} \left(\frac{T - T'}{TT'} \right)$$

$$\log(2.7296) = \frac{E_a}{2.303 \times 8.314} \left(\frac{308 - 298}{298 \times 308} \right)$$

$$E_a = \frac{2.303 \times 8,314 \times 298 \times 308 \times \log(2.7296)}{308 - 298}$$

$$= 76750.096 \text{ J mol}^{-1}$$

$$= 76.75 \text{ kJ mol}^{-1}$$

calculate k at 318 K

$$A = 4 \times 10^{10} \text{ s}^{-1}, T = 318 \text{ K}$$

Again, from Arrhenius equation, we get

$$\log = \log A - \frac{E_a}{2.303RT}$$

$$\log(4 \times 10) - \frac{76.75 \times 10^3}{2.303 \times 8.314 \times 318}$$

$$= 10.6021 - 12.6051 = -2.003$$

$$k = \text{Anti log}(-2.003)$$

$$= 9.93 \times 10^{-3} \text{ s}^{-1}$$

30. The rate of a reaction quadruples when the temperature changes from 293 K to 313 K. Calculate the energy of activation of the reaction if it does not change with temperature.

Ans: From Arrhenius equation, we get

$$\log, \log \frac{k_2}{k_1} = \frac{E_a}{2.303} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Given

$$k_2 = 4k_1$$

$$T_1 = 293\text{K}$$

$$T_2 = 313\text{K}$$

$$\text{So, } \log \frac{4k_2}{k_1} = \frac{E_a}{2.303 \times 8.314} \left(\frac{313 - 293}{293 \times 313} \right)$$

$$0.6021 = \frac{20 \times E_a}{2.303 \times 8.314 \times 293 \times 313}$$

$$E_a = \frac{0.6021 \times 2.303 \times 8.314 \times 293 \times 313}{20}$$

$$= 52863.00 \text{ J mol}^{-1}$$

$$= 52.86 \text{ kJ mol}^{-1}$$

As a result, the activation energy required is $52.86 \text{ kJ mol}^{-1}$